

Financial Time Series

TA Session #3

April 9, 2021

Check on definition of

$$\bar{\Phi}(B)$$

$$\Theta(B)$$

??

Slide 5

lecture 5

$$\begin{aligned}
 y_{n+1} &= \mu + \phi_1 y_n + \phi_2 y_{n-2} + \epsilon_{n+1} \\
 \hat{y}_{n+1} &= \hat{\mu} + \hat{\phi}_1 y_n + \hat{\phi}_2 y_{n-2} + 0 \\
 \hat{y}_{n+2} &= \hat{\mu} + \hat{\phi}_1 y_{n+1} + \hat{\phi}_2 y_{n-2} + 0
 \end{aligned}$$

MA2

$$\begin{aligned}
 y_n &= \mu + \epsilon_n + \epsilon_{n-1} \theta_1 + \epsilon_{n-2} \theta_2 \\
 \hat{y}_{n+1} &= \hat{\mu} + \hat{\epsilon}_{n+1} + \hat{\epsilon}_n \theta_1 + \hat{\epsilon}_{n-1} \theta_2 \\
 \hat{y}_{n+2} &= \hat{\mu} + \hat{\epsilon}_{n+2} + \hat{\epsilon}_{n+1} \theta_1 + \hat{\epsilon}_n \theta_2
 \end{aligned}$$